Quiz 10

Friday, December 4 — least squares, residuals, fitting, and positive definiteness

Last updated: August 18, 2026.

Related lectures: Least squares, Fitting.

Related lab: Lab 5.

Schedule

SVD is not on this quiz; it is introduced afterward. See the capstone packet for SVD practice.

Core

- B 12.1, 12.4, 12.10, 12.12
- B 13.1, 13.3, 13.6, 13.11, 13.14
- L Ch. 2 Ex. 7: least-squares line fit
- L Ch. 2 Ex. 8: simulated data and fitting
- C10.1: Compute and interpret a residual vector.
- C10.2: Compare two fits using RMS residual.
- C10.3: Given information about the columns of A, decide whether $A^\mathsf{T} A$ is positive definite and explain.

Stretch

B 12.2, 12.7, 13.4.

What a quiz item might change

One observation in a tiny dataset, a fitted line whose residuals you interpret, two RMS errors to compare, or why $A^\mathsf{T} A$ is positive definite when A has independent columns.